

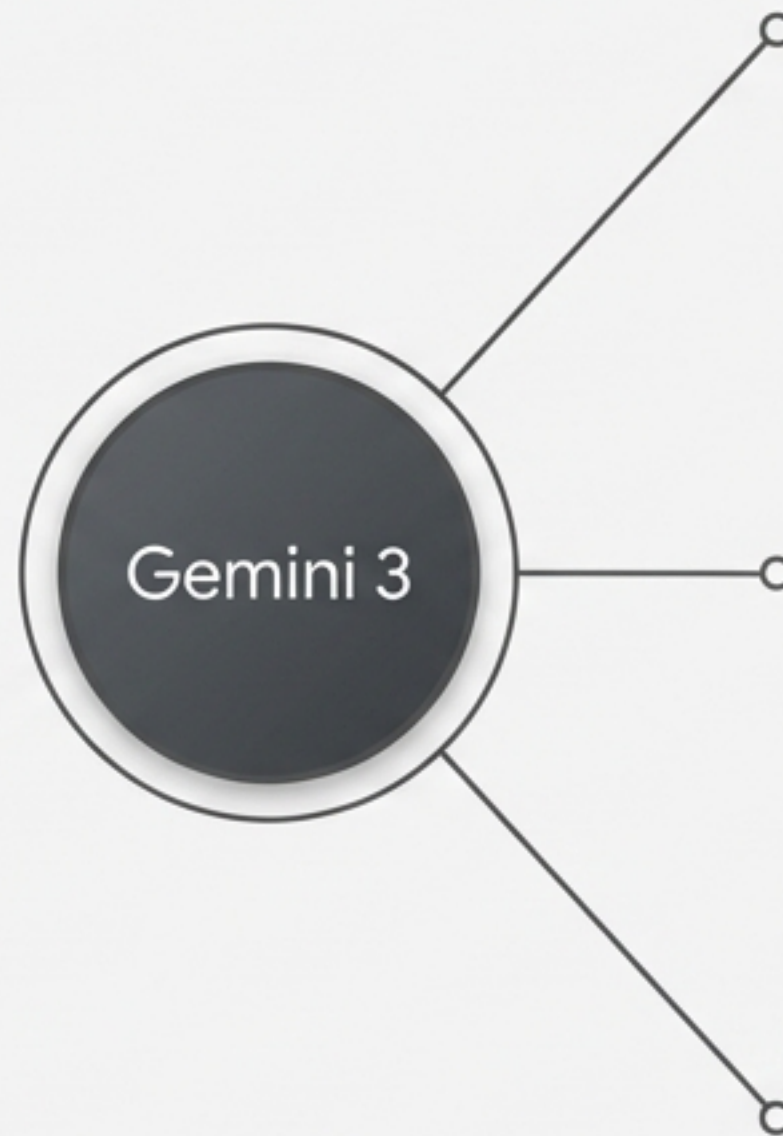
Gemini 3: A New Architecture for Intelligence

An Analytical Deep Dive into the State of the Art in Reasoning, Multimodality, and Agentic AI



More Than an Upgrade: A Fundamental Leap Forward

Google's Gemini 3 is not an incremental improvement. It represents a foundational shift in AI, setting a new state-of-the-art by unifying three core capabilities at an unprecedented scale. This deck provides a comprehensive analysis of this leap, organized around three pillars:



1. State-of-the-Art Reasoning: Achieving new benchmarks in complex problem-solving and introducing 'Deep Think' for unparalleled depth.



2. Natively Multimodal: Built from the ground up to seamlessly understand and generate across text, code, images, and video—not as separate skills, but as a unified intelligence.



3. The Agent-First Developer Experience: Empowering developers with advanced agentic coding, "vibe coding," and the new Google Antigravity platform to build more autonomous applications.

The Culmination of a Two-Year Scientific Endeavor

As articulated by CEO Sundar Pichai, Gemini 3 builds directly on the foundations laid by its predecessors, now shipping at the scale of Google.

.....
Gemini 1 —————● **Gemini 2** —————● **Gemini 3** —————→

The Foundation: Introduced breakthroughs in native multimodality and a long context window, expanding the *types* and *volume* of information an AI could process.

The Agent: Laid the groundwork for agentic capabilities and pushed the frontiers of reasoning, enabling the model to handle more complex, multi-step tasks. Topped LMArena for over six months with Gemini 2.5 Pro.

The Synthesis: Combines these capabilities into our most intelligent model, built to grasp depth, nuance, and intent to bring any idea to life.

Pillar 1: State-of-the-Art Reasoning with Unprecedented Depth

Gemini 3's reasoning is designed to go beyond surface-level answers. It grasps nuance, peels apart overlapping layers of difficult problems, and trades cliché for genuine insight. It's a true thought partner.

Key Advancement: Gemini 3 Deep Think

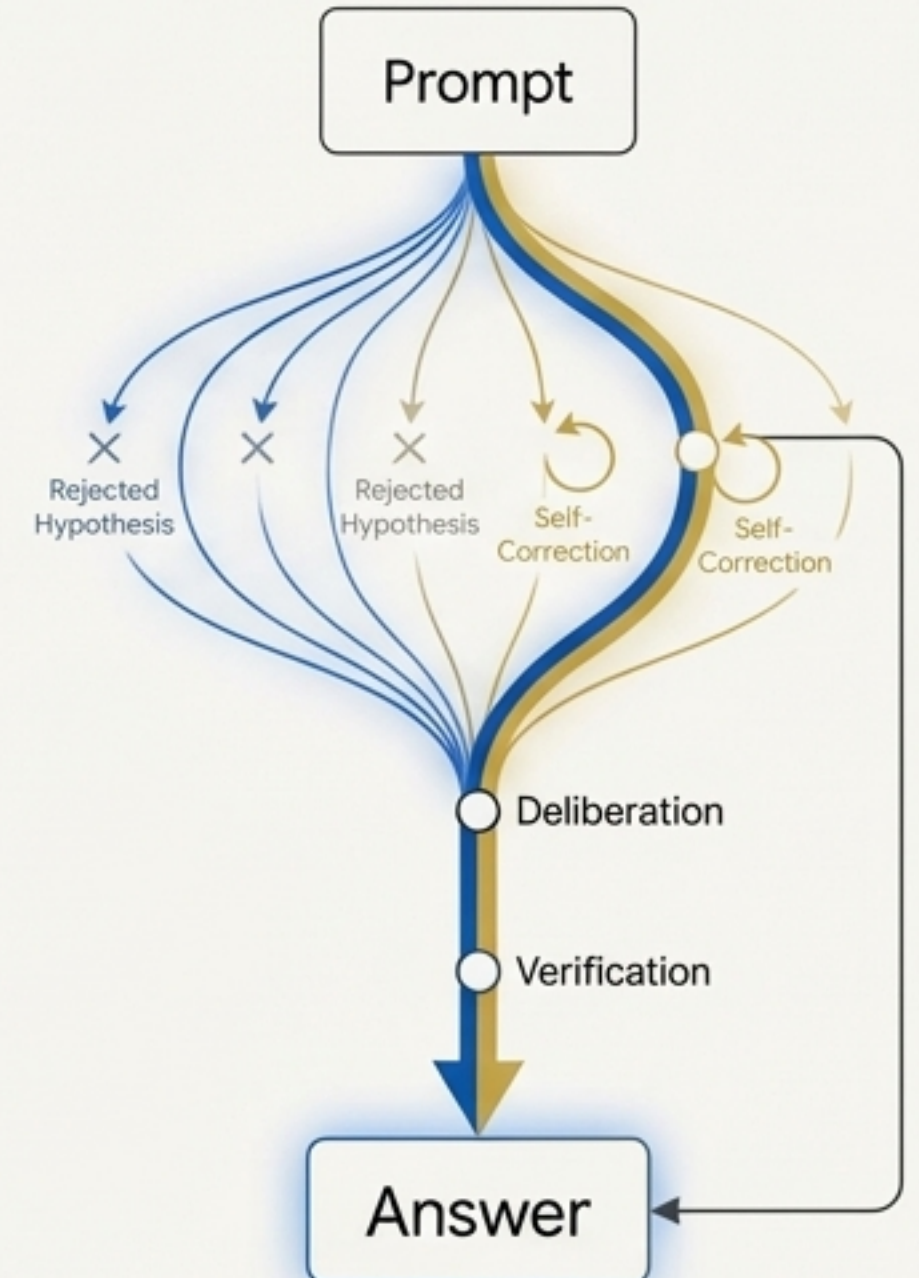
- * An enhanced reasoning mode that delivers a step-change in performance for the most complex problems.
- * It is not a separate model, but a configuration that allocates more internal compute, longer deliberation chains, and stricter verification heuristics before responding.
- * This mode trades higher latency for higher-quality, more deliberative outputs, designed for novel challenges where correctness is paramount.

“It's amazing to think that in just two years, AI has evolved from simply reading text and images to reading the room.”
– Sundar Pichai, CEO

Standard Reasoning



Deep Think Process



The Proof: Dominance Across Key Industry Benchmarks

Gemini 3 Pro significantly outperforms previous models and competitors across a wide range of academic and industry benchmarks, establishing a new state of the art.

Benchmark	Description		Gemini 3 Pro	Gemini 2.5 Pro	Claude Sonnet 4.5	GPT-5.1
Humanity's Last Exam	Academic reasoning	No tools	37.5%	21.6%	13.7%	26.5%
ARC-AGI-2	Visual reasoning puzzles	ARC Price Verified	31.1%	4.9%	13.6%	17.6%
GPQA Diamond	Scientific knowledge	No tools	91.9%	86.6%	83.4%	88.1%
AIME 2025	Mathematics	No tools With code execution	95.0%	88.0%	87.0%	94.0%
MathArena Apex	Challenging Math Contest problems		23.4%	0.5%	1.6%	1.0%
MMMU-Pro	Multimodal understanding and reasoning		81.0%	68.0%	68.0%	76.0%
ScreenSpot-Pro	Screen understanding		72.7%	11.4%	36.2%	3.5%
CharXiv Reasoning	Information synomation from complex charts		81.4%	69.6%	68.5%	69.5%
OmniDocBench 1.5	OCR	Overall Edlt Distance, lower is better	0.115	0.145	0.145	0.147
Video-MMMU	Knowledge acquiation from videos		87.6%	83.6%	77.8%	80.4%
LiveCodeBench Pro	Competitive cooding problems from Codeforces, IPC, and ICI	Elo Rating, higher is better	2,439	1,418	2,243	
Terminal-Bench 2.0	Agentic terminal coding	Terminus-2 agent	54.2%	32.0%	42.8%	47.6%
<u>SWE-Bench Verified</u>	Agentic coding	Single attempt	76.2%	59.6%	77.2%	76.3%
τ2-bench	Agentic tool use		85.4%	54.9%	84.7%	80.2%
Vending-Bench 2	Long-horizon agentic tasks	Net worth (mean), higher is better	\$5,478.16	\$573.64	\$3,838.74	\$1,473.43
FACTS Benchmark Suite	Held out internal grounding, MIM, search retrieval betchmarks		70.5%	63.4%	50.4%	50.8%
SimpleQA Verified	Parametric knowledge		72.1%	54.5%	29.3%	34.9%
MMMLU	Multilingual GGA		91.8%	89.5%	89.1%	91.0%
Global PIQA	Commensense reaoring across 100 Languages and Cultures		93.4%	91.5%	90.1%	90.9%
MRCR v2 (8-needle)	Long context performance	128k (average) 1M (pointwise)	77.0% 26.3%	58.0% 16.4%	47.1% not supported	61.6% not suppo

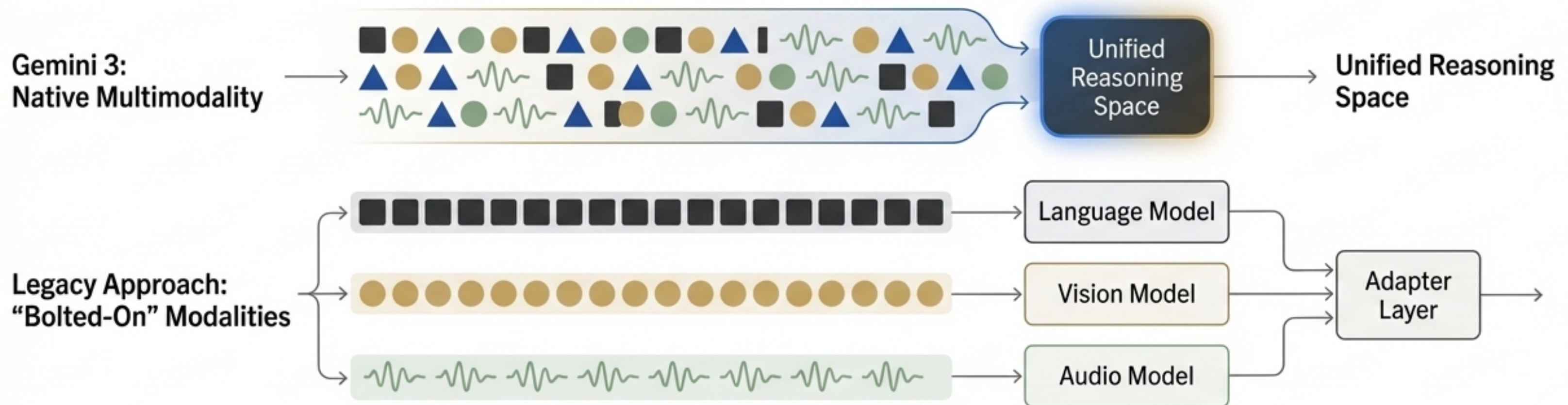
LMarena Leaderboard:
Breakthrough score of 1501 Elo

Pillar 2: Natively Multimodal from Day One

Unlike models that bolt on vision capabilities to a text-based core, Gemini was trained from the start to be natively multimodal. It doesn't translate images into text; it reasons in a unified high-dimensional space where different modalities coexist.

How it Works

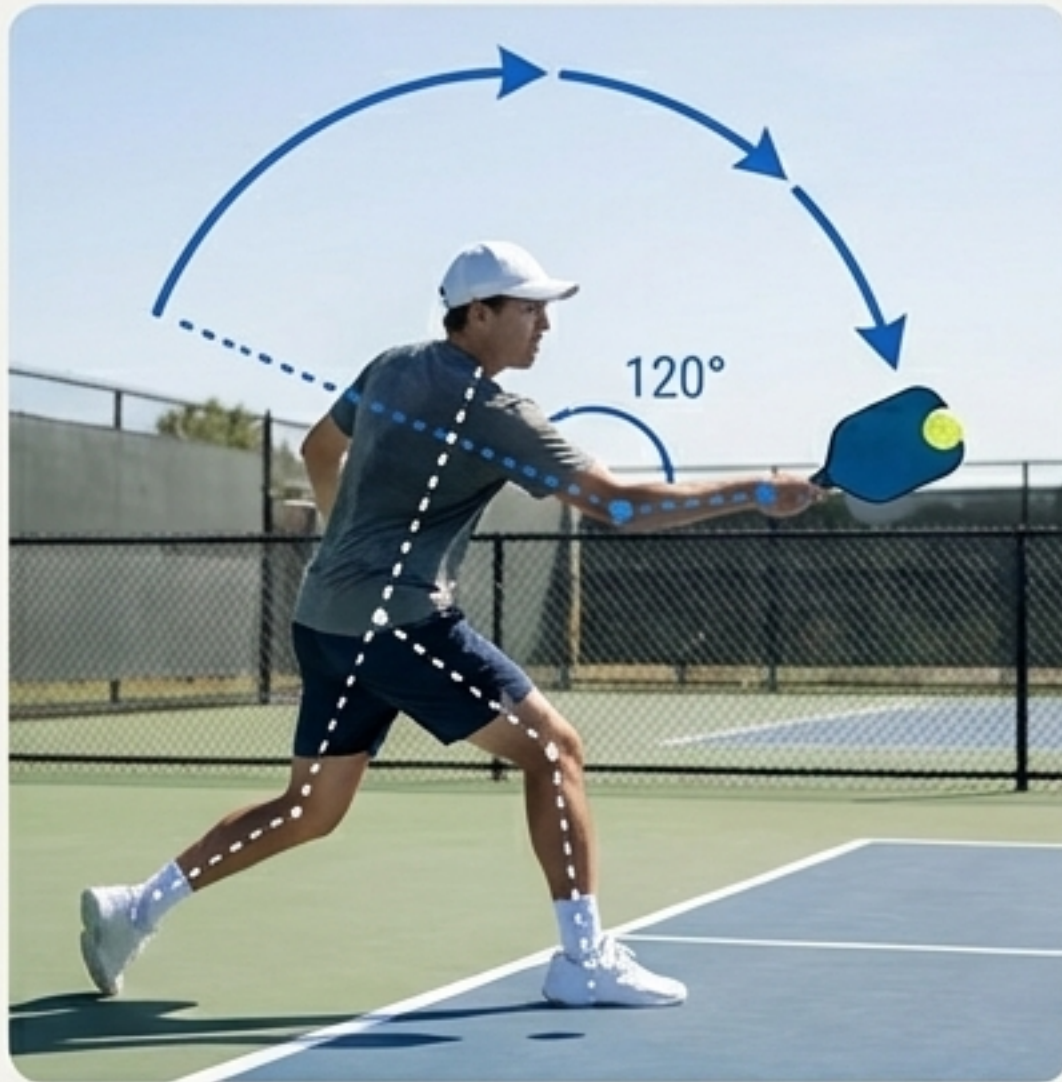
- **Interleaved Tokenization:** Gemini processes images, text, audio, and video not in sequence, but as a single, interleaved stream of tokens. This allows it to, for example, read code while simultaneously analyzing a corresponding UI diagram.
- **Dynamic Tiling for Vision:** Instead of downsampling high-resolution images and losing detail, images are divided into 768x768 tiles, which are then independently tokenized to preserve fidelity.
- **1 Million Token Context Window:** Enables processing of entire codebases, academic collections, or long videos in a single pass.



Multimodality in Action: From Pickleball Analysis to Playable Games

Gemini 3's native multimodality unlocks capabilities that go far beyond simple image description.

Learn Anything



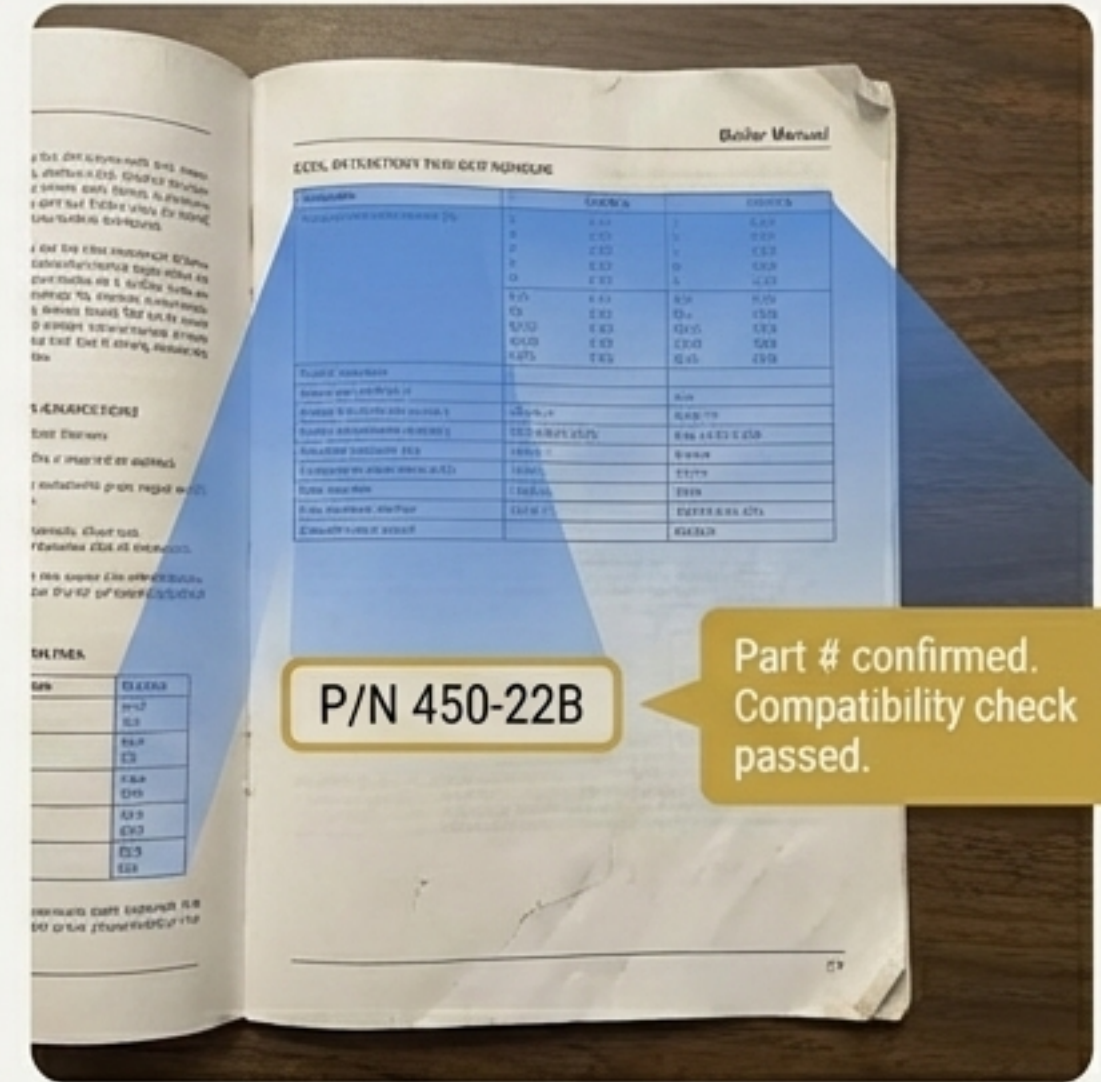
Analyze video of a pickleball match to identify areas for improvement and generate a training plan. Decipher and translate handwritten recipes from different languages to create a shareable cookbook.

Build Anything



Generate the complete code for a playable retro 3D spaceship game from a text prompt, including richer visualizations and interactivity.

Plan Anything



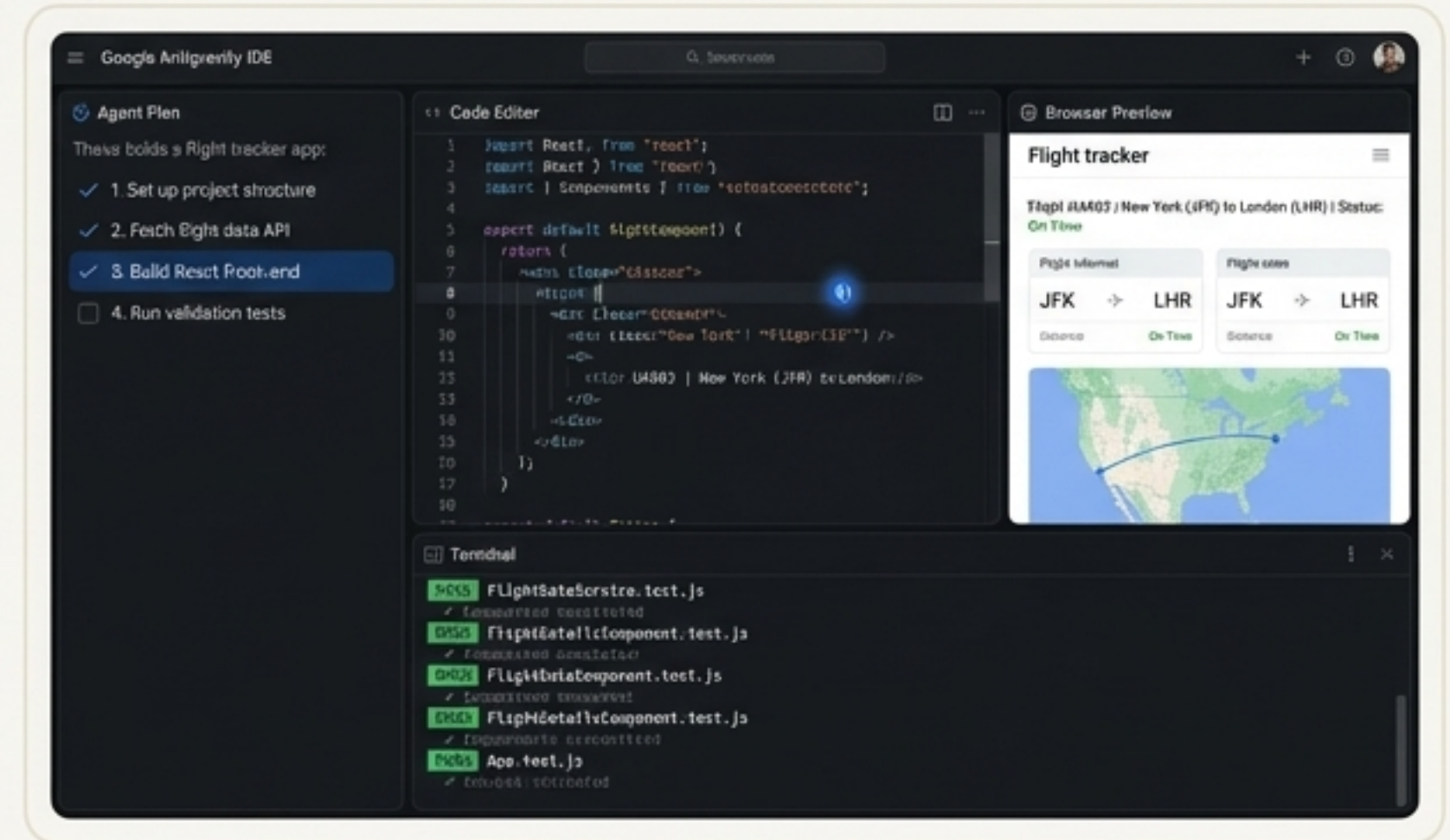
Parse a photo of a boiler manual to identify the correct part number and check for compatibility issues, as a user on Reddit successfully did.

Pillar 3: An Agent-First Developer Experience

Gemini 3 advances AI assistance from a tool *in* the developer's toolkit to an active partner. It excels at “vibe coding”—transforming high-level ideas into functional code—and agentic coding for autonomous task completion.

Introducing Google Antigravity

- A new agentic development platform that allows developers to operate at a higher, task-oriented level.
- Agents have direct access to the editor, terminal, and browser, enabling them to autonomously plan and execute complex, end-to-end software tasks.
- Leverages Gemini 3's advanced reasoning and tool use to independently code an application and validate its execution.

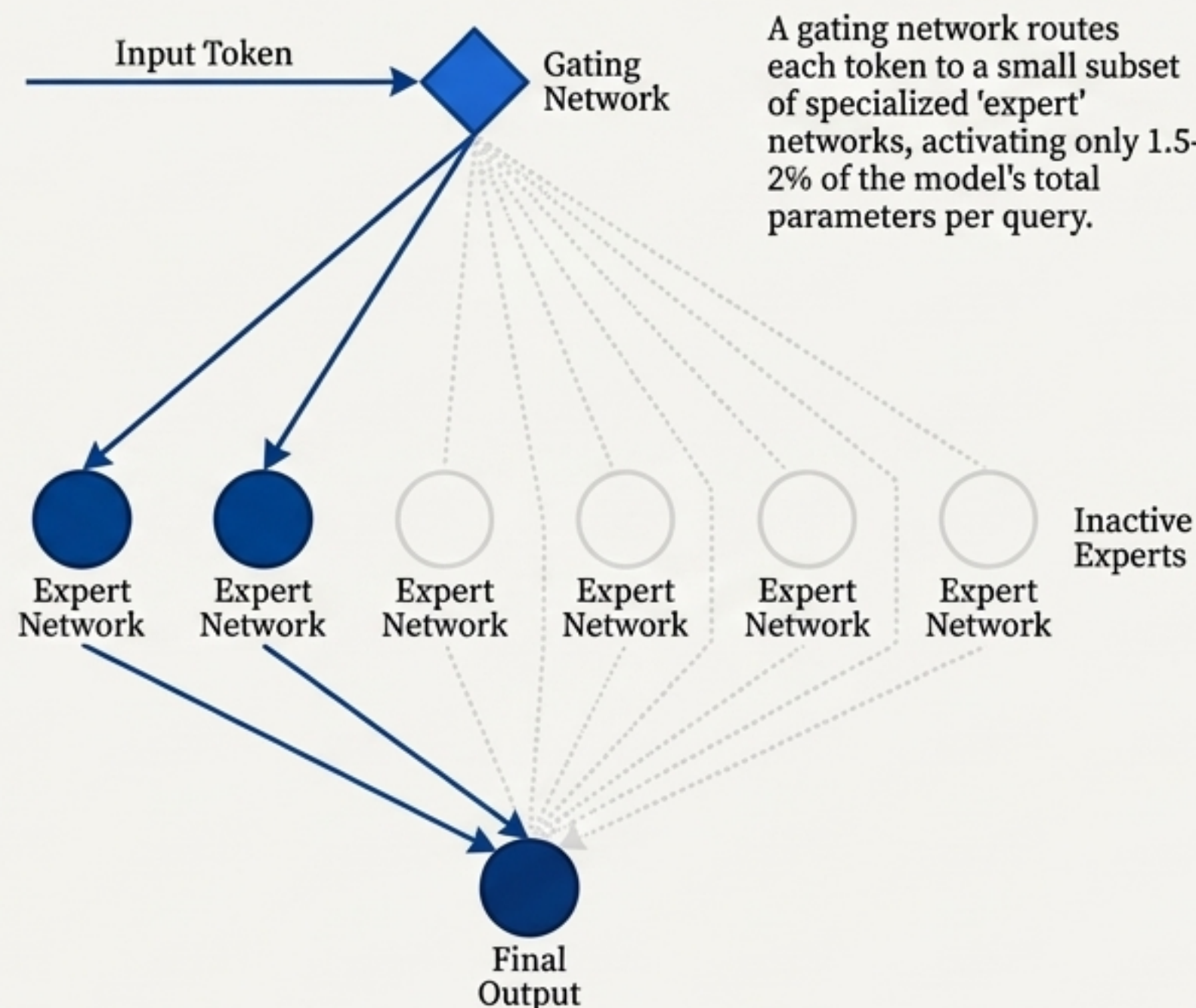


The Architecture Underpinning Gemini 3

Gemini 3 utilizes a Mixture of Experts (MoE) architecture, the mainstream approach for frontier models, to balance immense scale with computational efficiency.

Core Technical Specifications

- *** Architecture****: Mixture of Experts (MoE)
- *** Total Parameters****: Over 1 trillion (1T+)
- *** Activated Parameters****: 15-20 billion activated per query, reducing latency and cost compared to dense models.
- *** Context Window***: A massive 1 million token context window, offered in two tiers (200k and 1M) for flexibility. This is equivalent to processing a 300-page book or an entire codebase in one go.
- ***Key Improvement***: Implements stronger **structured planning and self-correction design** in the reasoning pipeline.
- ***Knowledge Cutoff***: January 2025 (API) / October 2025 (Analysis report).



For Developers: Mastering the New API Controls

Gemini 3 introduces new parameters giving developers granular control over performance, cost, and fidelity.

`thinking\_level`

Controls the **maximum** depth of the model's internal reasoning.

- **low**: Minimizes latency and cost for simple tasks.
- **high (Default)**: Maximizes reasoning depth for complex problems.

```
response = model.generate_content(  
    "How does AI work?",  
    config={  
        "thinkingConfig": {"thinkingLevel": "low"}  
    }  
)
```

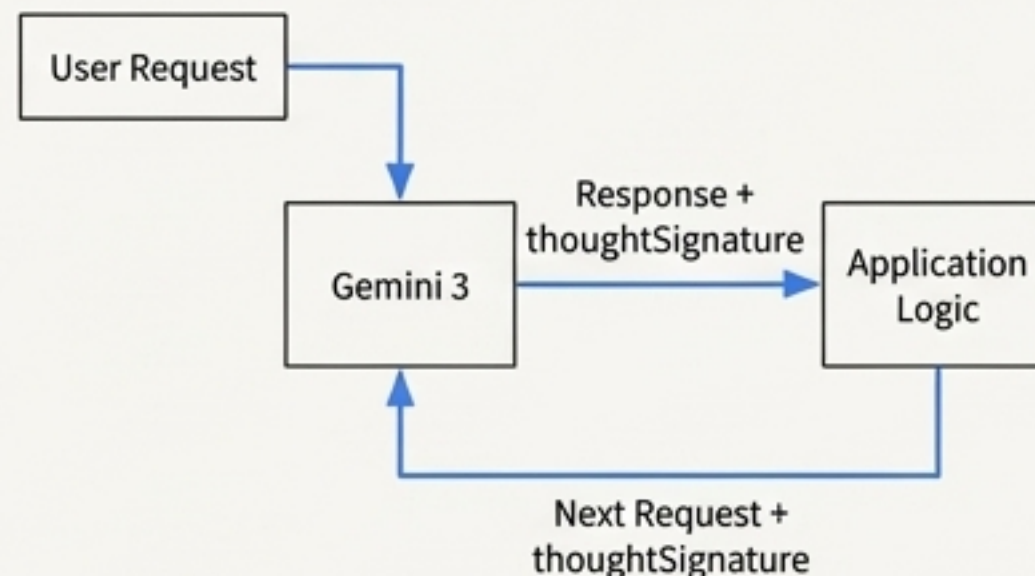
`media\_resolution`

Determines the maximum tokens allocated per image or video frame, allowing a trade-off between detail and cost. Critical for optimizing OCR on dense documents or video analysis.

```
response = model.generate_content([  
    "Describe this diagram.",  
    Part.from_uri(  
        "gs://image.jpg", "image/jpeg",  
        media_resolution={  
            "level": "media_resolution_high"  
        }  
    )  
)
```

`thoughtSignatures`

Crucial for agentic workflows. These are encrypted representations of the model's internal thought process. They **must** be returned in subsequent API calls to maintain reasoning context, especially for multi-step function calling. The official SDKs handle this automatically.



We strongly recommend keeping the **`temperature`** parameter at its default of 1.0. Lowering it may degrade reasoning performance.

The Gemini 3 Model Family: Pro, Flash, and Nano Banana

Model ID	Key Use Case	Context Window (In / Out)	Pricing (Input / Output per 1M tokens)
gemini-3-pro-preview	Complex tasks, advanced reasoning, broad world knowledge.	1M / 64k	Tiered: \$2/\$12 (<200k), \$4/\$18 (>200k)
gemini-3-flash-preview	Pro-level intelligence at high speed and lower cost. Best for high-throughput apps.	1M / 64k	\$0.50 / \$3
nano-banana-pro-preview (Gemini 3 Pro Image)	Highest quality image generation. Conversational editing.	65k / 32k	\$2 (Text In) / \$0.134 (Image Out)*

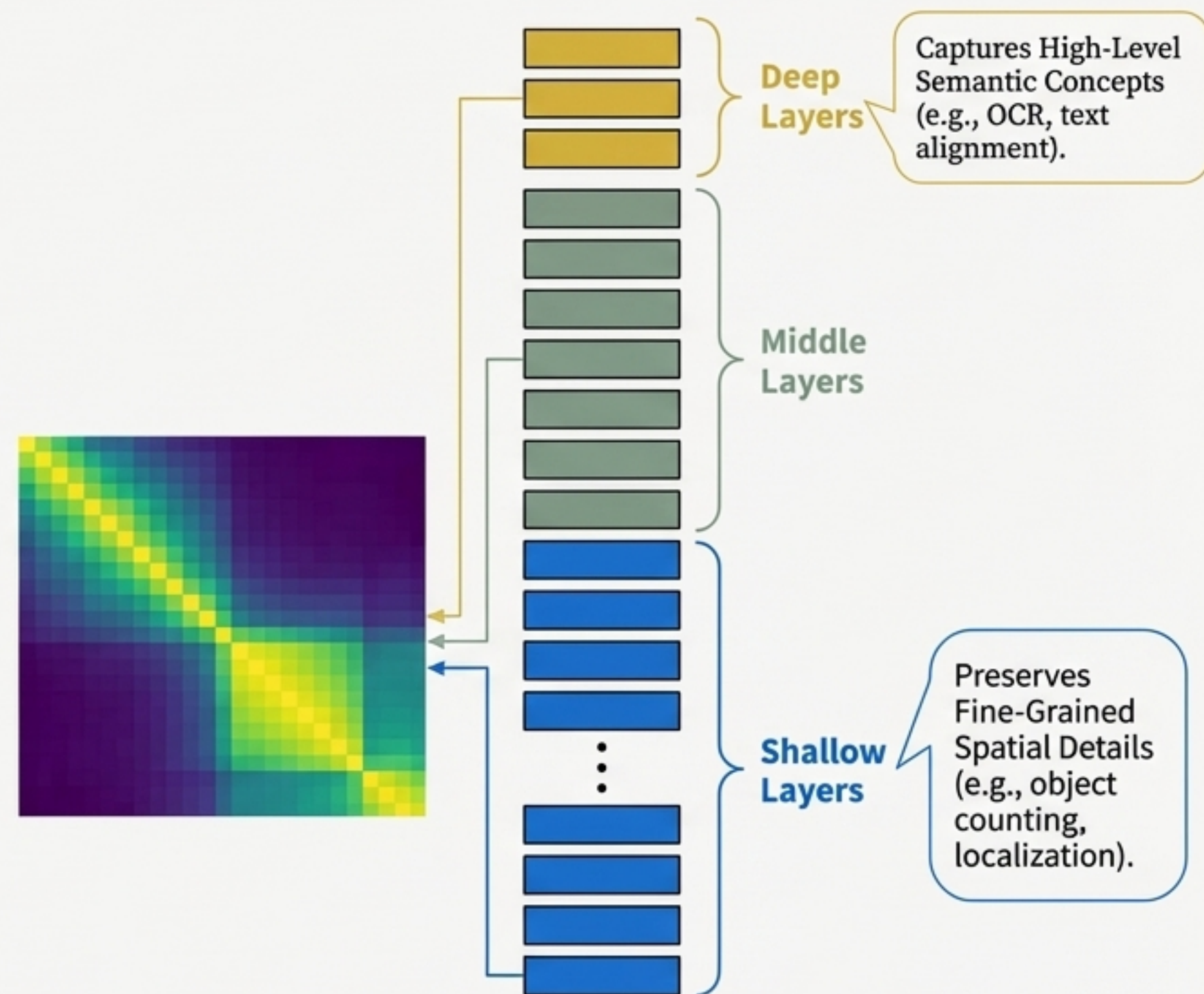
Knowledge cutoff for all models is January 2025. Image pricing varies by resolution. Full details on the pricing page.

Behind the Curtain: MLLMs See Better When They Look Shallower

Recent research (Chen et al., arXiv) reveals that vision in MLLMs is not monolithic. The common practice of using only the deepest layers of a Vision Transformer (ViT) is not always optimal.

- **Deep Layers (e.g., 21-24):** Excel at tasks requiring high-level semantic understanding, like OCR, because they are most aligned with textual concepts.
- **Shallow & Middle Layers (e.g., 1-20):** Significantly outperform deep layers on fine-grained visual tasks like counting, positioning, and object localization. They preserve crucial spatial details that can be suppressed in the final layers.

Key Takeaway: The penultimate layer (23) often represents the best balance, but for tasks demanding extreme visual precision, shallower features hold untapped value. This insight informs more sophisticated approaches to visual feature fusion.



The Reality Check: How the Community is Using (and Critiquing) Gemini 3

Real-world performance is the ultimate benchmark. Here's a look at early community reception from developers and enthusiasts.

Praise: Where It Excels

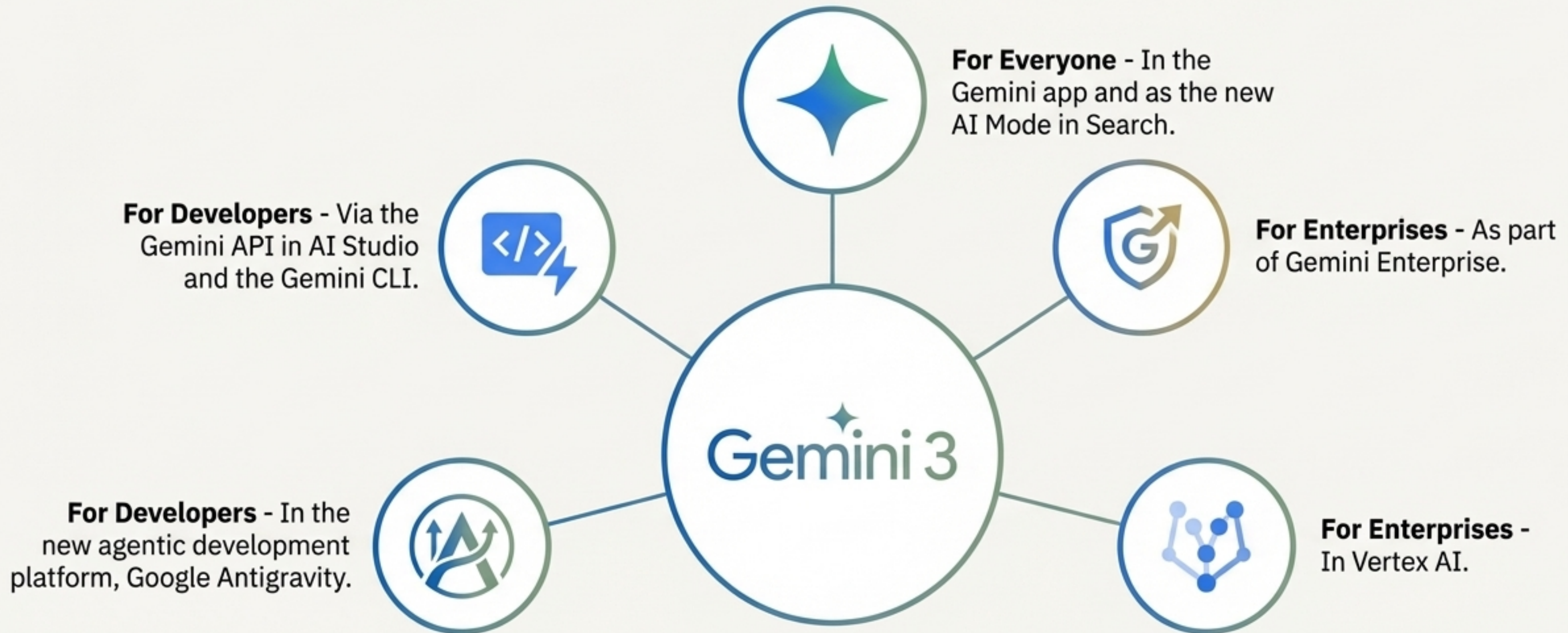
- “**Complex Coding:** “The coding agent is just SHITTING on GPT5 in my personal use on an extremely complex project.”
- “**One-Shot Solutions:** “I bit the bullet and signed up for free trial on Gemini and it legitimately 1-shot the solution [for a complex DAX problem].”
- “**Creative Generation:** “New Gemini is the only one to be able to achieve [making a fully stylized playable backgammon game].”
- “**Practical Multimodality:** “It read my boiler manual accurately enough to spot a flue kit was the wrong size... unlike gpt5.”

Critiques: The Current Limitations

- “**Instruction Following:** “I am leaning towards so far that chatgpt 5.1 is better at instruction following.”
- “**Reasoning Loops:** “Gemini on the other hand got caught in a loop of ‘Yes, sir, it’s fixed now’ and reproducing the same problem.”
- “**Accuracy Issues:** “It couldn’t fix a simple bug that involved just 1 missing line of - 1 word code after 2 attempts. ChatGPT did it in 1 attempt.”
- “**‘Training to the Test’:** “Because it’s only good at benchmarks and tests that investors look at... For the end user, chatbots are plateauing.”

A Fully Integrated Ecosystem

Gemini 3 is not just an API; it's the new intelligence layer being deployed across Google's entire stack, creating a cohesive ecosystem for users, developers, and enterprises.



The Dawn of the Gemini 3 Era

The launch of Gemini 3 marks the beginning of a new chapter in AI development. The convergence of state-of-the-art reasoning, native multimodality, and an agent-first development platform points to a future where AI is less of a tool and more of an autonomous collaborator.

By pushing the frontiers of intelligence, agents, and personalization, this new architecture is designed to enable applications that can learn, build, and plan with a depth and nuance previously out of reach. The era of truly helpful, multimodal AI is here.

